From Data To Insights: Digital Strategies for Automotive

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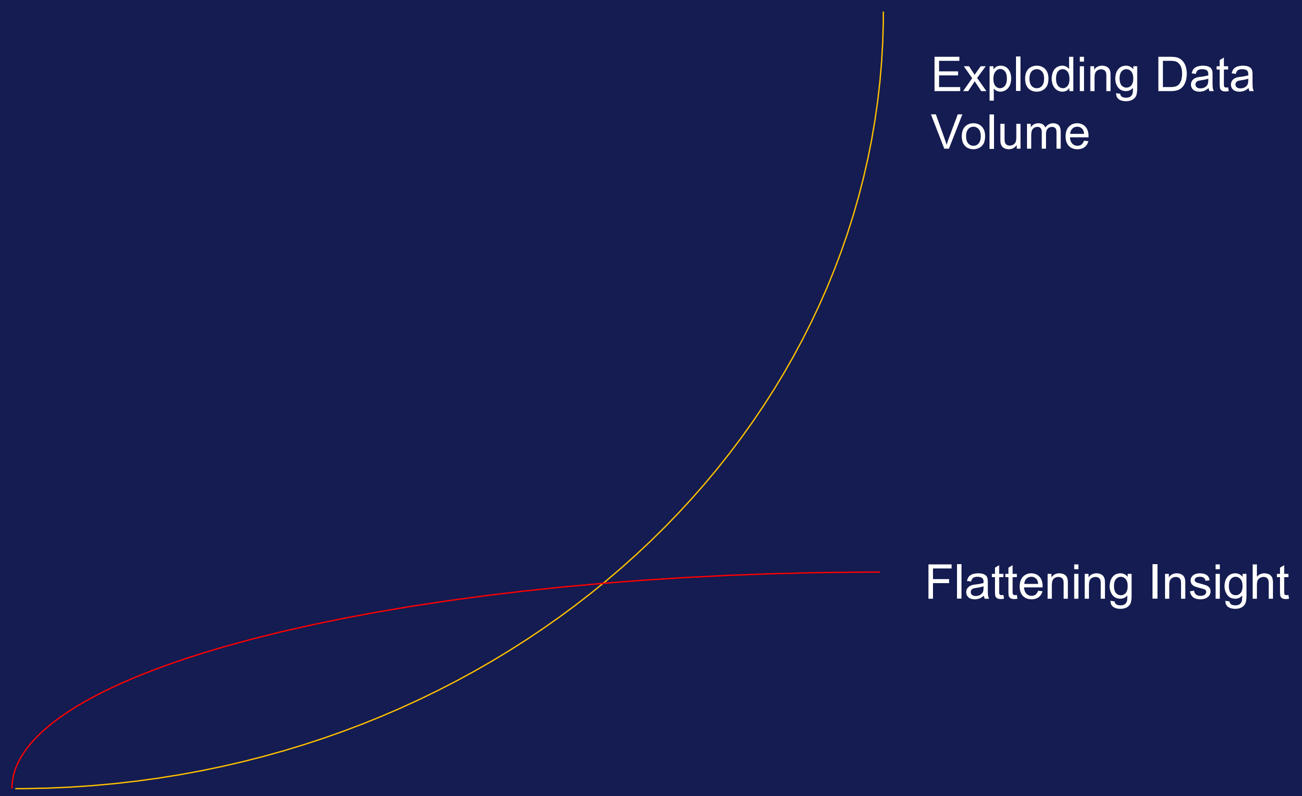
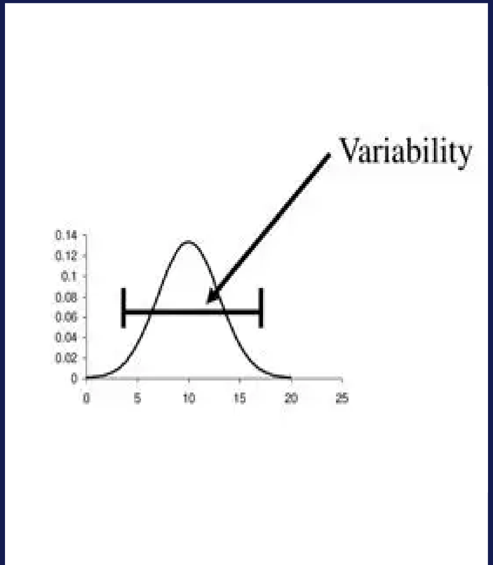
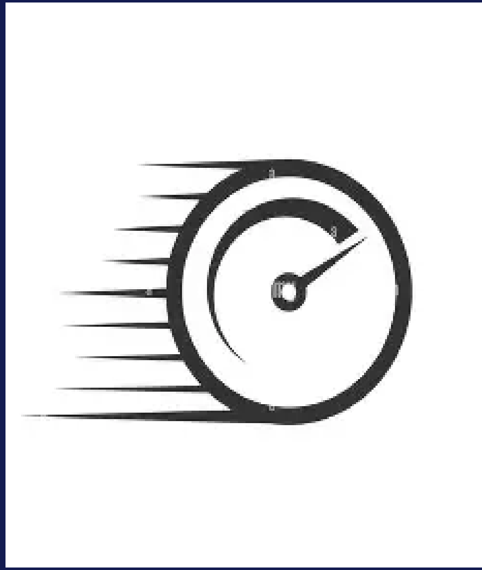
Key Challenges In
Automotive Testing
Safety-Criticality
Complexity of Systems
Integration Testing
Regulatory Compliance
Real-World Environment Simulation
Data Security and Privacy
Software Updates and Maintenance
Time-to-Market Pressure
Cost Constraints
Talent Shortage

We have more Data than ever – and less Clarity!



THE HIDDEN FRICTION IN AUTOMOTIVE DEVELOPMENT…





Disconnected Processes, Tools

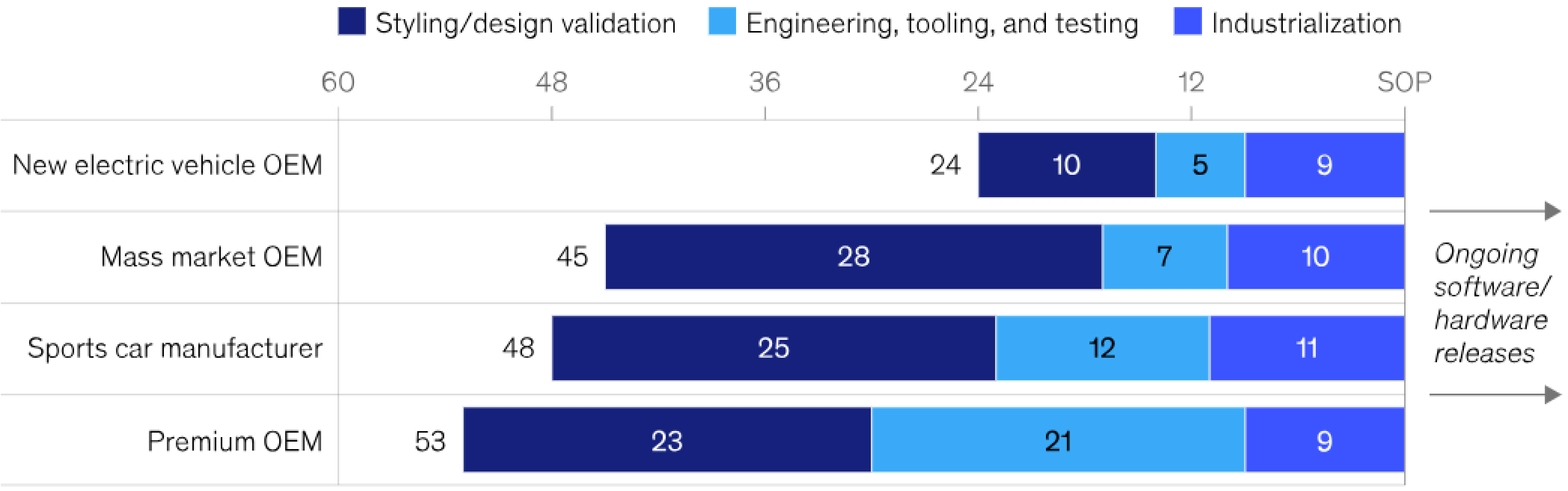
Silos between Functions

High Variability

Lack of Speed

New electric vehicle OEMs develop products in just 24 months—up to twice as fast as China's other automakers.

Directional product development timeline, months to start of production (SOP)



Source: Expert interviews; McKinsey analysis

McKinsey & Company

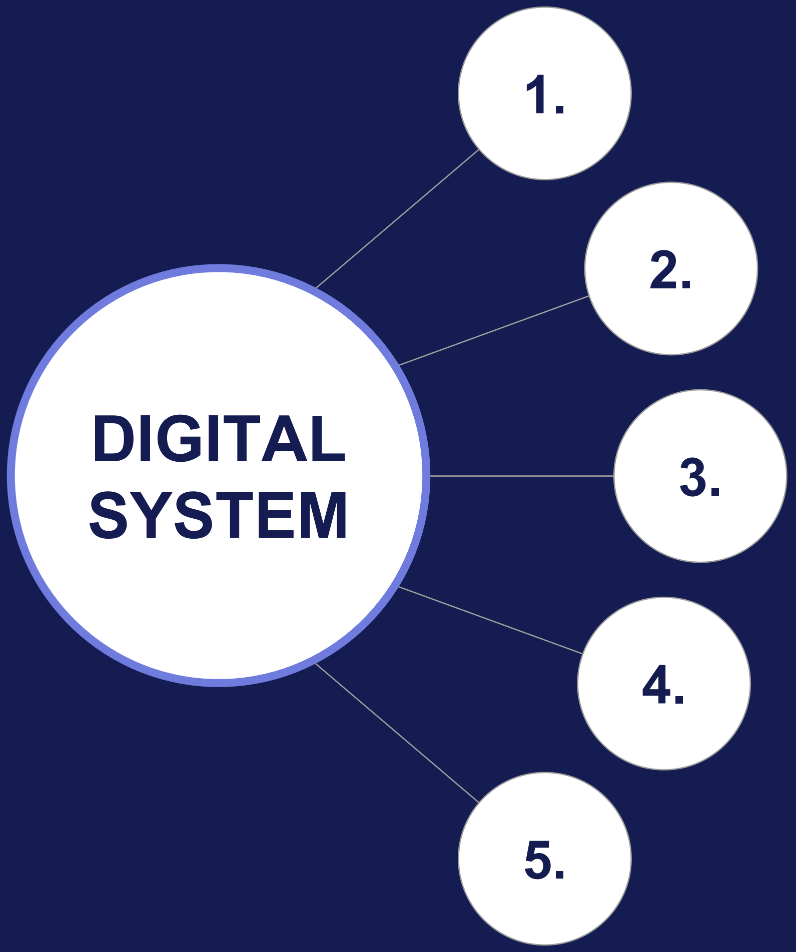
How they do it…

Estimated reduction in Time to Market (in months)

https://www.mckinsey.com/capabilities/operations/our-insights/automotive-product-development-accelerating-to-new-horizons

Automotive Development is now a Digital System problem.

THE DIGITAL PLAYBOOK



Design for Data

Unify Physical and Virtual

Automate Insight Extraction

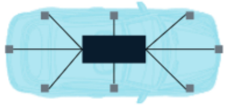
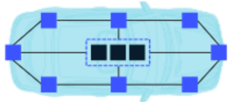
Close the Loop Early

Achieve Traceability & Trust

EXAMPLE: SOFTWARE DEFINED VEHICLES

Several archetypes are likely to bring more centralized control in generations four and five.

■ ECU[1] ■ DCU[2] ■ Zone computer ■ Central computer

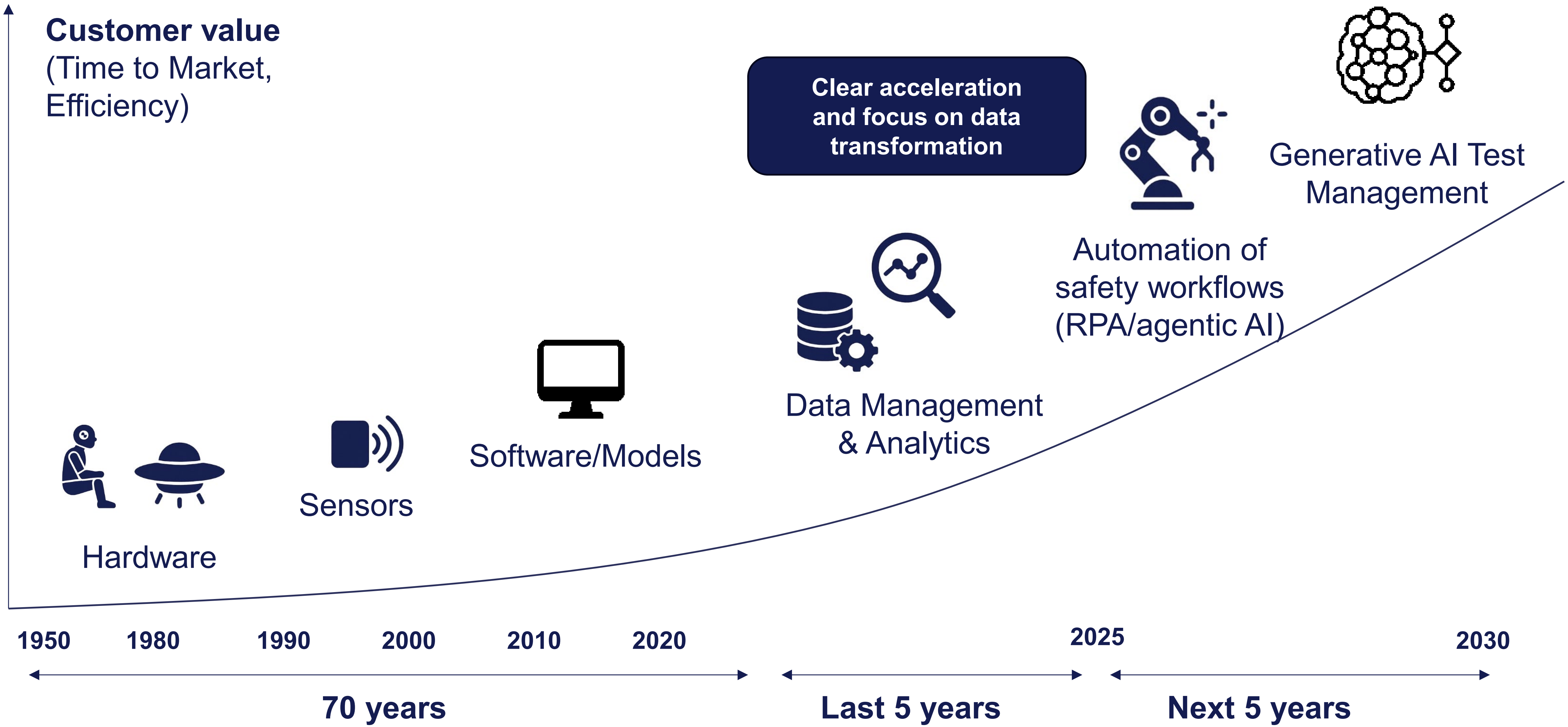
Generation	Archetype	Description	
5th generation: Vehicle centralized	Central brain	Powerful central general-purpose processing unit with mostly unprocessed inputs from electronic control units (ECUs) Optimized for camera and deep-learning-based approach to highly autonomous driving	
	Zone computing	Majority of domains controlled by central virtual domain units based on a general-purpose compute cluster Backed by distributed-zone computers across the vehicle Best for OEMs with limited number of platforms and variants	
4th generation: Domain centralized	Full-domain centralization	Domain-control-unit (DCU) architecture used for all domains Well suited for multiple platforms Suitable for level-3 autonomous driving	
	Selected-domain centralization	Selective domains using DCUs (typically starting with infotainment and ADAS[3]) Good for cost minimization	

[1]Electronic control systems.
[2]Domain control systems.
[3]Advanced driver-assistance systems.

McKinsey & Company

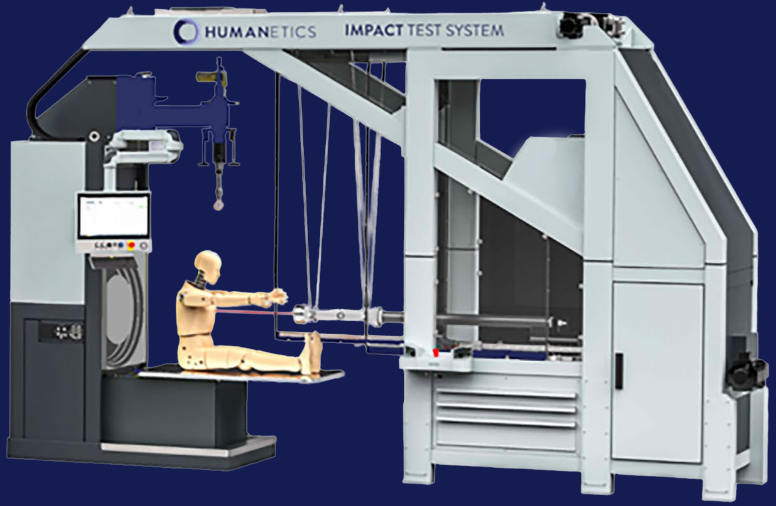
1. **Design for Data:** Early architecture decisions
2. **Unify Physical & Virtual:** Software enables HW
3. **Automate Insight Extraction:** Incident Analytics
4. **Close the Loop Early:** Early ‘in loop’ Testing
5. **Trust & Traceability:** Traceability built in

HUMANETICS: FROM HARDWARE TO INTELLIGENCE PLATFORM BUSINESS

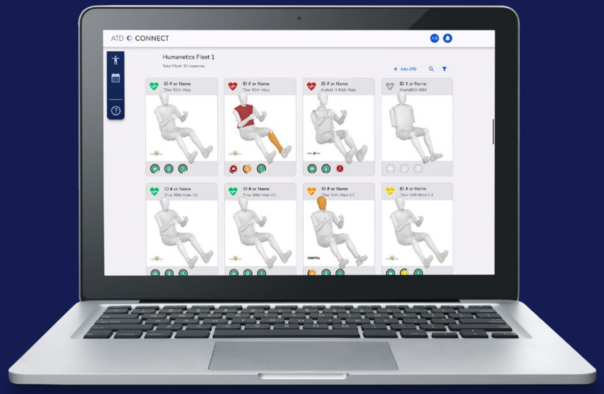


HUMANETICS

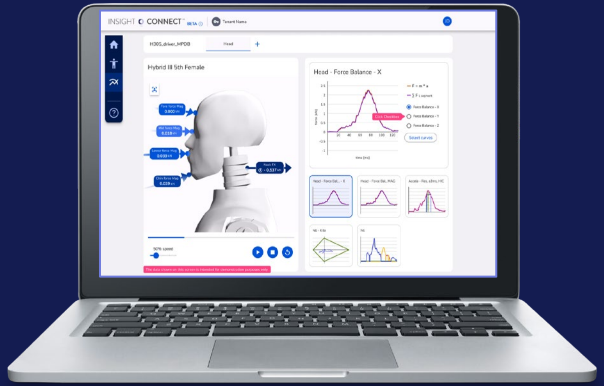
HUMANETICS SAFETY INTELLIGENCE

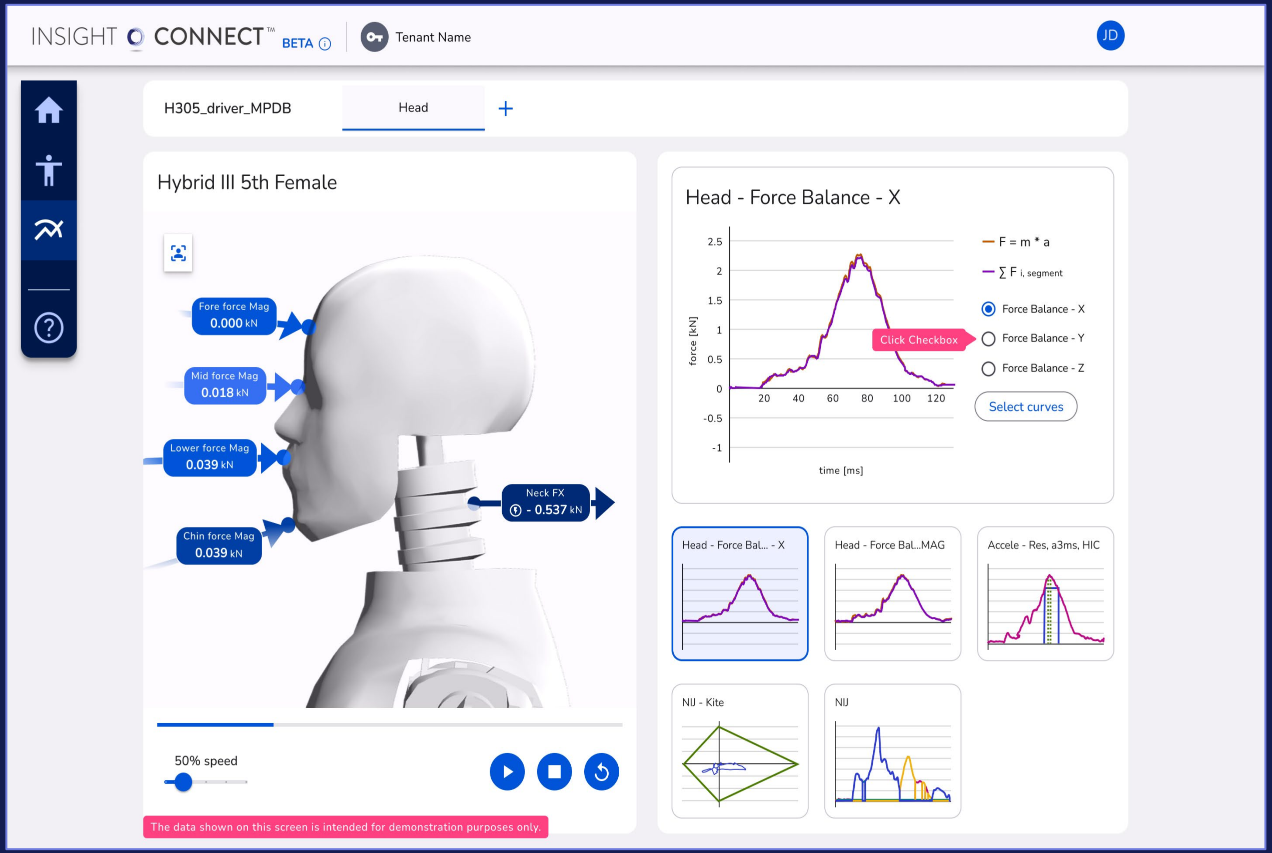


Digital ATD Lab



HUMANETICS SAFETY INTELLIGENCE

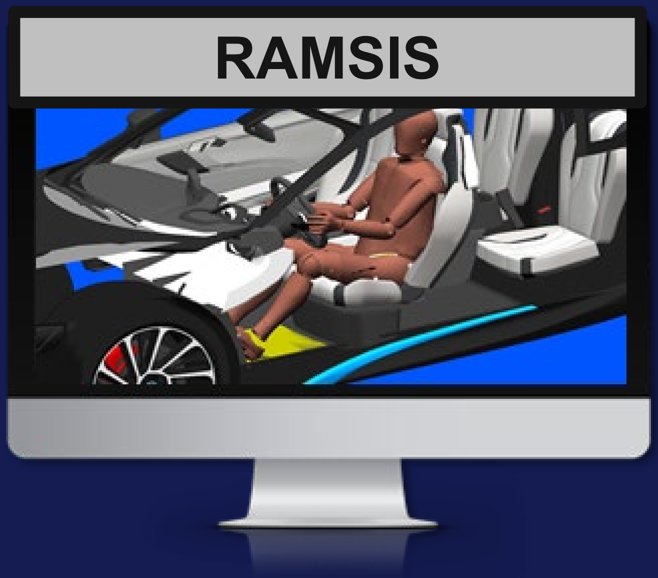


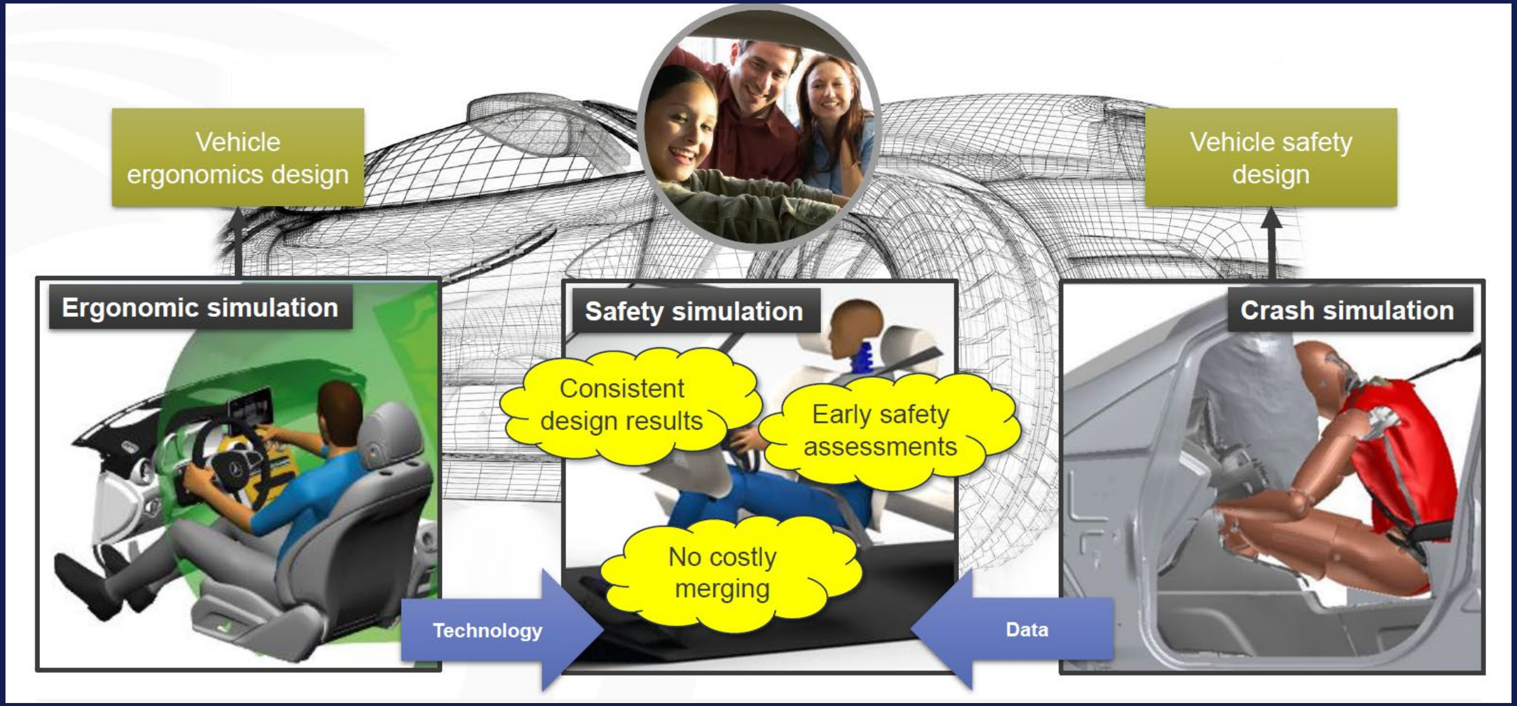


ATD Simulation Insights

HUMANETICS

HUMANETICS COMFORT INTELLIGENCE





Human Factors Insights and Automation

EMPOWERING HUMAN FACTORS DESIGN…

HOW CAN WE COLLABORATE?

HUMANETICS

Thank You